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QUIZ 1

Given that two functions (**funA** and **funB**) have been defined as follows:

```
void funA(int n) {
  cout << "funA - " << n << "\n";
  n = n + 1;
  if (n < 5) {
    if (n % 2 == 0) {
      funB(n);
    } else {
      funA(n);
    }
  }
}
```

```
void funB(int n) {
  cout << "funB - " << n << "\n";
  n = n + 1;
  if (n < 5) {
    if (n % 2 == 0) {
      funB(n);
    } else {
      funA(n);
    }
  }
}
```

Study the funA and funB functions above. Identify the followings:

1. Instruction to check if the terminal case has been reached: <u>if (n < 5)</u>

2. Instruction to change the size of the problem so the terminal case can be reached: <u>n = n + 1</u>

Trace the recursive calls of the funA and funB functions inside the following main function of a C++ program:

```cpp
int main() {
  funA(1);
  return 0;
}
```

funA(1);

```cpp
void funA(1){
    cout << "FunA - " << n << "\n";
    n = 1 + 1;
    if (2 < 5){
        if (n % 2 == 0){
            funB(2);
        }
        else{
            funA(n);
        }
    }
}
```

Output :
FunA - 1

```cpp
void funB(2){
    cout << "FunB - " << n << "\n";
    n = 2 + 1;
    if (3 < 5){
        if (n % 2 == 0){
            funB(n);
        }
        else{
            funA(3);
        }
    }
}
```

Output :
FunA - 1
FunB - 2

```cpp
void funA(3){
    cout << "FunA - " << n << "\n";
    n = 3 + 1;
    if (4 < 5){
        if (n % 2 == 0){
            funB(4);
        }
        else{
            funA(n);
        }
    }
}
```

Output :
FunA - 1
FunB - 2
FunA - 3

```cpp
void funB(4){
    cout << "FunB - " << n << "\n";
    n = 4 + 1;
    if (5 < 5){
        if (n % 2 == 0){
            funB(n);
        }
        else{
            funA(3);
        }
    }
}
```

Reach Terminal Case

Output :
FunA - 1
FunB - 2
FunA - 3
FunB - 4

Final Output :
FunA - 1
FunB - 2
FunA - 3
FunB - 4